BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 024 · Data structures: OCR exam practice

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. When an item is popped from a stack, is the oldest or the most recently pushed item removed first?

Q2. Queue: which item is removed first?

Q3. What do left/right columns contain in an array-based BST?

Q4. What four actions does OCR 1.4.2(c) require?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — LIFO

Recall the term — FIFO

Recall the term — pointer

Recall the term — record

Recall the term — tuple

Recall the term — BST

Recall the term — collision

New learning

Stack wording — CODE:
stack → LIFO
push → add to top
pop → remove from top — OCR questions often test both the operation and the consequence.

Queue wording — CODE:
queue → FIFO
enqueue → rear
dequeue → front — State the ordering difference clearly rather than saying only 'they work differently'.

Linked-list wording — CODE:
start pointer → first item
next pointer → index of next item
null → end — OCR June 2023 explicitly tests how the second item is accessed using pointer values.

BST table wording — CODE:
Left | Data | Right
pointer | value | pointer — OCR examiners reported that candidates often incorrectly put data rather than pointer indexes in left/right columns.

Worked example

Turn understanding into a 2-mark comparison

1. Name stack rule — CODE:
Stack removes the most recently added item first.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. Name queue rule — CODE:
Queue removes the earliest added item first.

3. Link the two — CODE:
Therefore the same input order produces different removal orders.

4. Use exam terms — CODE:
LIFO versus FIFO

We Do · Practise with guidance

Array vs list

Give one clear difference.

Improve the answer

```
An array and a list store multiple values, but ...
```

```
An array has a fixed size once created, whereas a list can grow or shrink.
```

Tuple vs list

State one difference.

```
A tuple differs because ...
```

```
A tuple is fixed / cannot be changed in the same way, whereas a list is
  mutable and can change.
```

Linked-list pointer

Explain how to reach the second item.

```
Use the ? pointer to locate the first item, then ...
```

```
Use the start pointer to locate the first item, read its next pointer, then
  use that pointer value as the index of the second item.
```

BST pointer table

What belongs in left/right columns?

```
Left/right contain ...
```

```
Left/right contain pointer/index values to child rows; leaf pointers are null.
```

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Compare structures
Answer the four short questions comparing stack/queue, array/list, tuple/list, and linked-list pointer access.

Question 2

BST array representation
Complete the Left/Data/Right representation for the Irish towns BST.

Question 3

Record + BST
Explain why a record suits a task containing an ID, real value and order number; then state two BST characteristics.

Question 4

Create / traverse / add / remove
For linked list, stack, queue, graph, BST and hash table, state how one add or remove operation works.

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

In OCR's June 2024 BST table, the left and right columns contain pointer/index values rather than copies of town names.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The most recently added item (LIFO).

Q2. The earliest added item (FIFO).

Q3. Pointers/indexes.

Q4. Create, traverse, add data, remove data.

You Do pages · Independent questions

Q1. Compare structures
CODE:
Stack vs queue:
A stack is LIFO; a queue is FIFO.

Array vs list:
An array has a fixed size once created; a list can grow/shrink.

Tuple vs list:
A tuple is fixed/immutable in structure whereas a list can be changed.

Linked-list second item:
Use the start pointer to locate the first item. Read that item's next pointer and use its value as the index/address of the second item.
Check: Use precise structure vocabulary and relational wording.

Q2. BST array representation
CODE:
0: 1 | Sligo | 3
1: 2 | Dublin | 4
2: null | Cork | null
3: 6 | Waterford | null
4: 7 | Galway | 5
5: null | Limerick | null
6: null | Tralee | null
7: null | Dundalk | null

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Check: Use pointer indexes in Left/Right and null for leaf pointers.

Q3. Record + BST
CODE:
Record:
It groups several related fields about one task into a single structured item, while allowing the fields to have different data types.

BST characteristics:
Each node has at most two children.
Values/keys smaller than a node are placed to the left and larger values/keys to the right.
Check: Link each answer directly to the context.

Q4. Create / traverse / add / remove
CODE:
Linked list add: update pointers so the new item's next points to the following item.
Stack add/remove: push/pop at the top.
Queue add/remove: enqueue at rear, dequeue at front.
Graph add/remove: set/clear adjacency-matrix entry or entries.
BST add: follow comparisons until a null child pointer and attach the new index.
Hash table add: hash the key, then place it in the bucket using the collision strategy.
Check: Your responses describe the structural change, not just name the structure.

Exit Ticket · Model answer

True. In a pointer-based representation of a BST, left and right fields identify child nodes, not copies of the node labels.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding